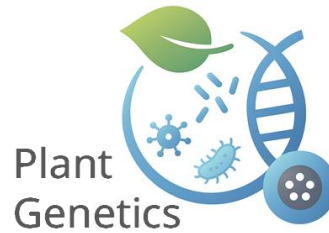


Microbiome Data analysis in Agriculture

Lecturer Assistant
Tian Tian

Main content

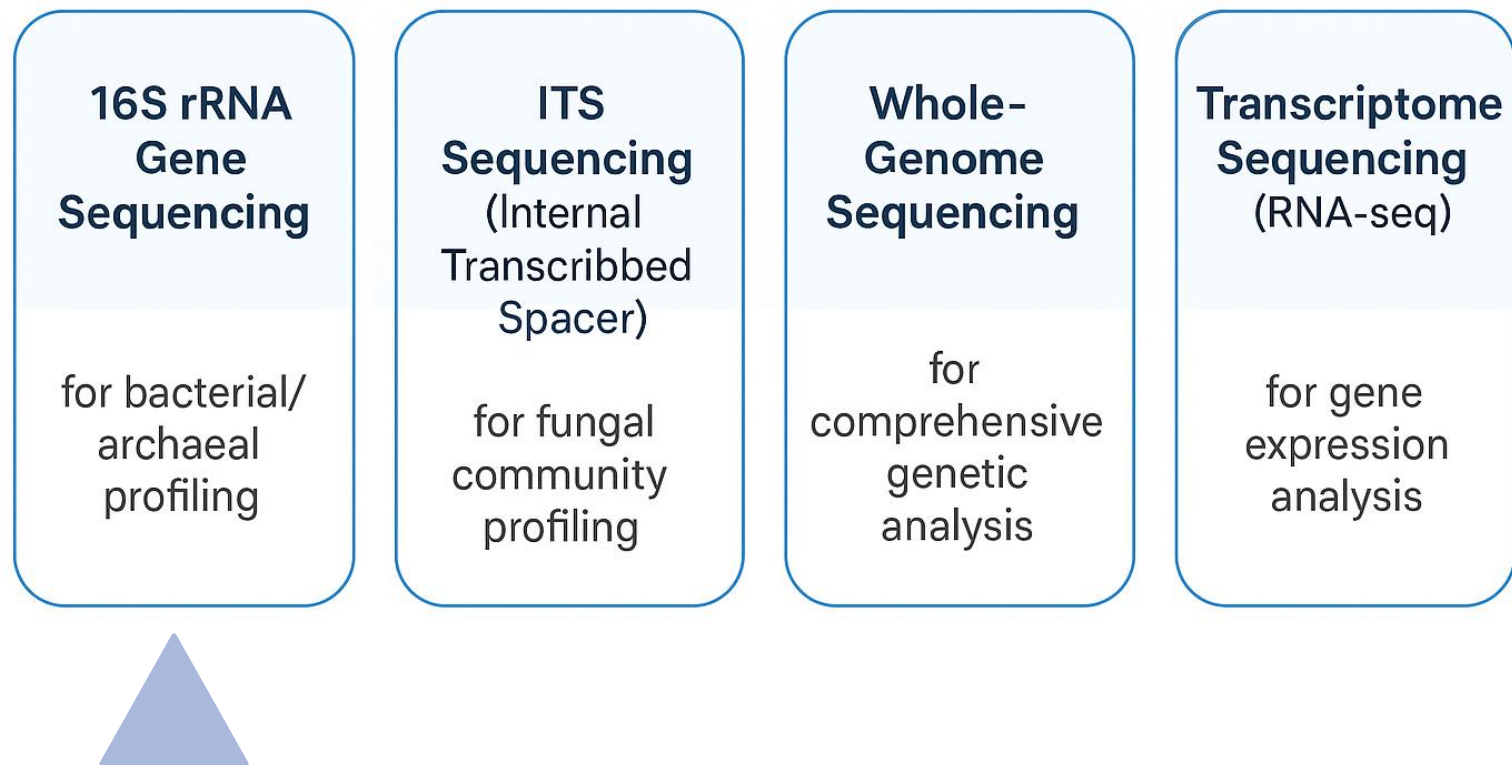
- I Omics Data in Root – Microbe Studies (Nov. 24)
- II Key Statistical Metrics in Microbiome Analysis (Dev. 1)
- III Root transcriptome data analysis (Dev. 8)
- IV Gene Expression – Microbiome Co-regulation Networks (Dev. 15)



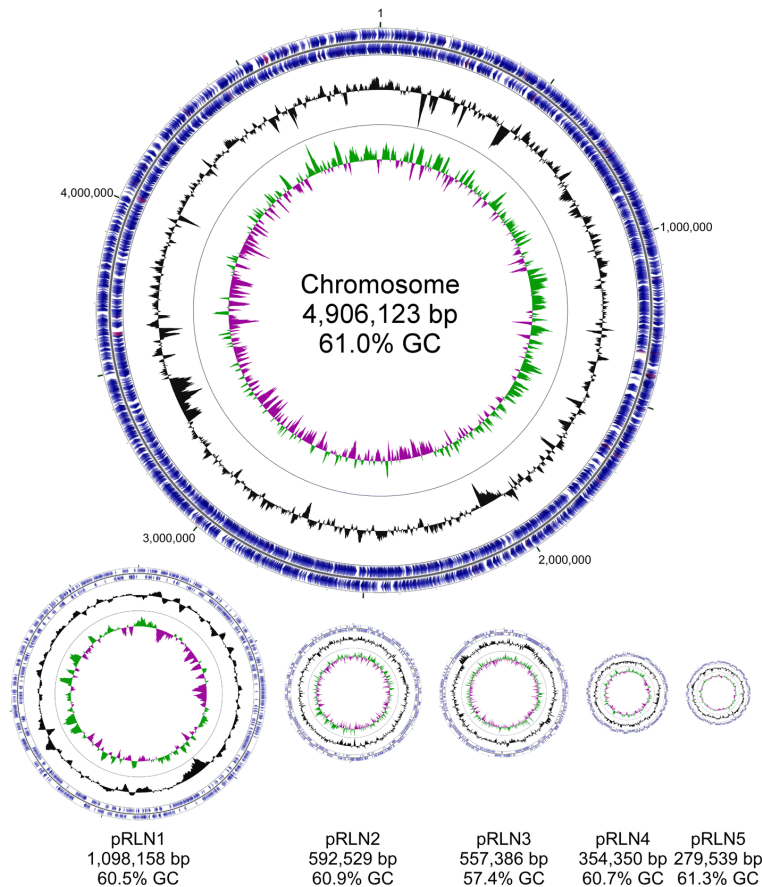
I Omics Data in Root – Microbe Studies (11.24)

Omics Data in Root–Microbe Studies

- **Research objective:** To understand the main types of omics data relevant to microbiome research.



Bacterial genome



➤ Key Features of a Bacterial Genome:

- Structure: Single circular chromosome (0.5–10 Mb) + multiple small plasmids.
- Size: Highly variable (0.5–10 Mb).
- Assembly: Requires strain-level sequencing.
- Classification: 16S rRNA gene as key marker.

Explore bacterial genomes:



MaGe platform: <https://mage.genoscope.cns.fr>

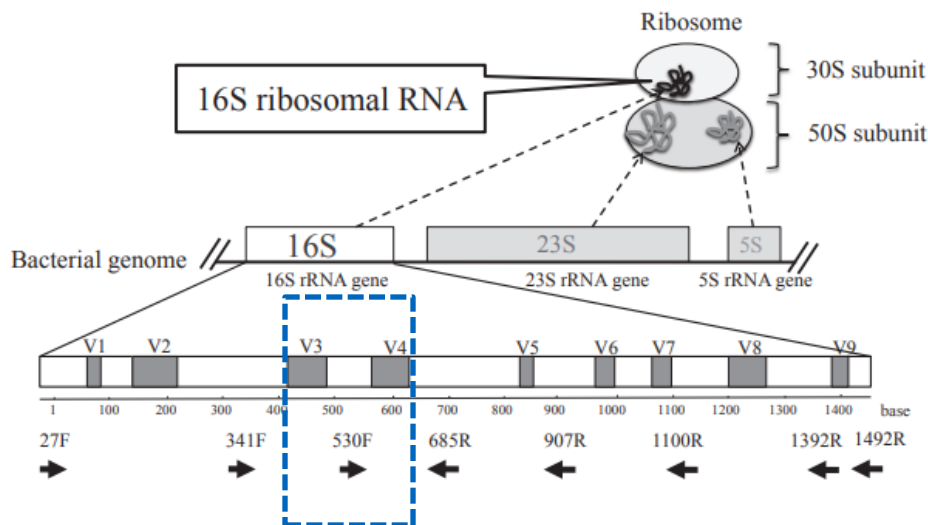
The chromosome and five plasmids of RI Norway.

(Liang et al., 2018, Environmental Microbiome)

16S rRNA gene

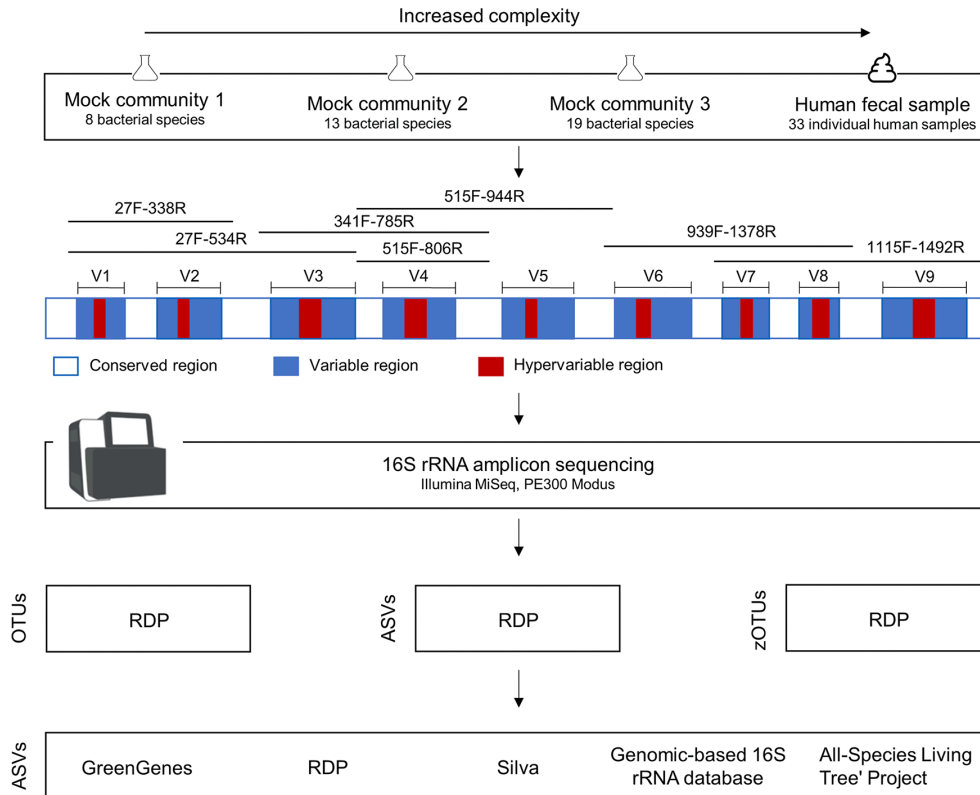
➤ Key Features:

- Highly conserved across all bacteria
- Contains variable regions for species discrimination
- ~1,500 bp length (9 hypervariable regions)
- Slow evolutionary rate → stable for taxonomy



(Fukuda et al., 2016, J UOEH)

16S rRNA gene amplicon sequencing



■ Typical Primers in Plant Root Microbiome Research

■ 515F-806R

- 5' GTGCCAGCMGCCGCGGTAA 3'
- 5' GGACTACHVGGGTWTCTAAT 3'
- Amplification length ~250bp

■ OTU (Operational Taxonomic Unit)

- [ASV \(Amplicon Sequence Variant\)](#)
- [A Higher-Resolution Alternative to OTU Clustering](#)

(Abellan-Schneyder et al., 2021,mSphere)

16S rRNA gene amplicon sequencing

1. Understanding Raw Sequencing Data

Assess sequencing quality using tools like FastQC or MultiQC.

2. Reads Alignment / Denoising

Clean and denoise reads (e.g., via DADA2, Deblur) or align to reference genomes.

3. Taxonomic Classification

Assign taxonomy using reference databases (e.g., SILVA, Greengenes).

4. Quantification

Generate abundance tables for taxa or functional genes.

Preparation for the upcoming exercises

- References
- Test datasets
- Softwares

References

The ISME Journal (2021) 15:3181–3194
<https://doi.org/10.1038/s41396-021-00993-z>



ARTICLE



➤ Genome wide association study reveals plant loci controlling heritability of the rhizosphere microbiome

Siwen Deng^{1,2} · Daniel F. Caddell · Gen Xu^{3,4} · Lindsay Dahlen^{1,5} · Lorenzo Washington¹ · Jinliang Yang^{3,4} · Devin Coleman-Derr

Received: 27 May 2020 / Revised: 2 April 2021 / Accepted: 20 April 2021 / Published online: 12 May 2021
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nature
plants

ARTICLES

<https://doi.org/10.1038/s41477-021-00897-y>



➤ Plant flavones enrich rhizosphere Oxalobacteraceae to improve maize performance under nitrogen deprivation

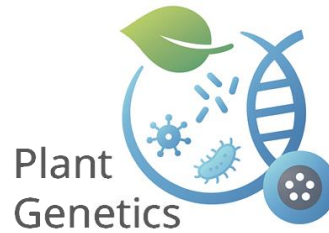
Peng Yu^{1,2,3,17}, Xiaoming He^{1,2,3,17}, Marcel Baer², Stien Beirinckx^{4,5,6}, Tian Tian⁷, Yudelsy A. T. Moya⁸, Xuechen Zhang⁹, Marion Deichmann , Felix P. Frey², Verena Bresgen^{2,3}, Chunjian Li¹¹, Bahar S. Razavi¹², Gabriel Schaaf , Nicolaus von Wirén , Zhen Su⁷, Marcel Bucher , Kenichi Tsuda , Sofie Goormachtig^{4,6}, Xinping Chen and Frank Hochholdinger

Test datasets

Data type	Download link
16S sequencing raw data	SRR11839631 SRR11839632 SRR11839633 SRR11839634 SRR11839635 SRR11839636 SRR11839637 SRR11839639 SRR11839640
OTU matrix	https://github.com/colemanderr-lab/Deng-2020/blob/master/fig1_24line.rds
RNA-seq raw data	SRR11839278 SRR11839289 SRR11839301 SRR11839312 SRR11839497 SRR11839508 SRR11839619 SRR11839630 SRR11839780
Gene expression matrix	https://github.com/PengYuMaize/Yu2021NaturePlants/blob/master/gene_counts_table_WG_CNA.txt
Reference genome	https://download.maizegdb.org/Genomes/B73/Zm-B73-REFERENCE-NAM-5.0/

Softwares

- Rstudio/R
 - Conda
 - **sratoolkit (conda install -c bioconda sra-tools)**
 - Hisat2
 - stringtie
 - samtools
-
- Install the appropriate version based on your system (Windows, macOS, or Linux).
 - <https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions>



II Key Statistical Metrics in Microbiome Analysis

(12.1)

Key Statistical Metrics in Microbiome Analysis

- **Research objective:** To understand effective traits through the analysis of microbiome data.
 - **Alpha diversity**
 - **Rarefaction analysis**
 - **Beta diversity**
 - **Taxonomy distribution**
- **Code:** https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/tree/main/01_16S

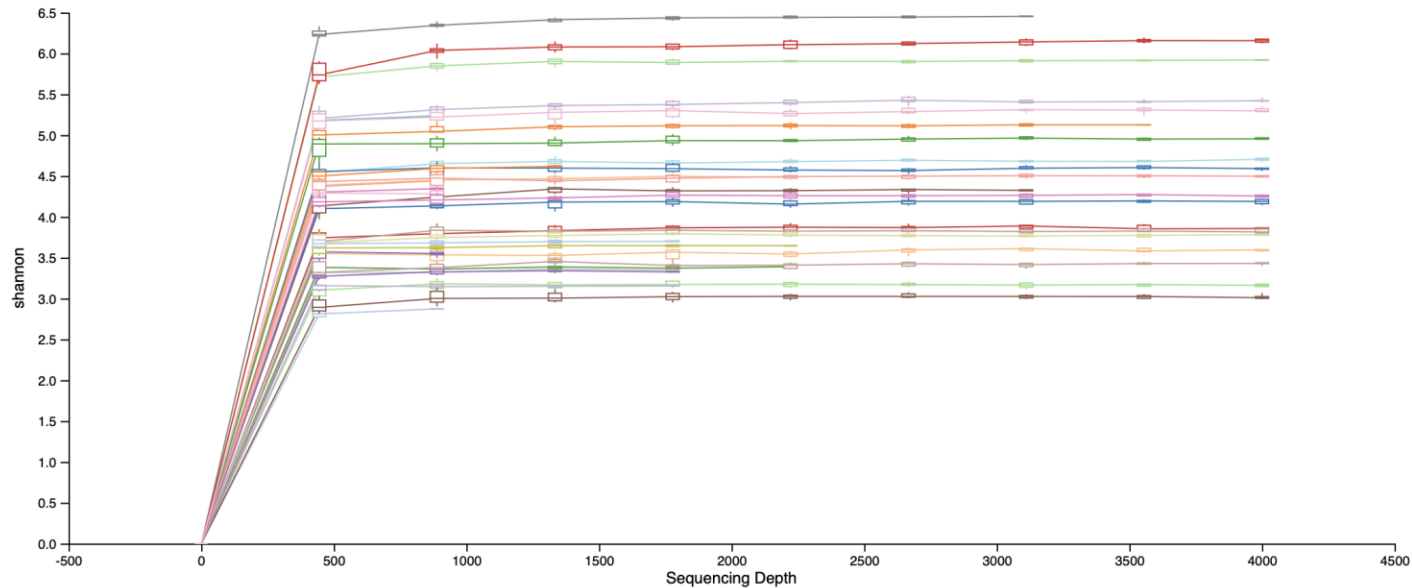
Alpha diversity

- Alpha diversity is within sample diversity.
- species diversity
- species evenness

Diversity Index	Description
Shannon Index	Considers both species richness and evenness; sensitive to rare species.
Simpson Index	Measures the probability that two randomly selected individuals belong to different species; more sensitive to dominant species.
Fisher's Alpha	Diversity index based on fitting a log series distribution; good for complex communities.

Rarefaction analysis

- Rarefaction curve illustrates how the cumulative number of alpha diversity metrics in a sample changes as sequencing depth increases.



From qiime2

Beta diversity

- Beta diversity is between sample diversity.
- It quantifies the dissimilarity or similarity in species composition between two or more samples.

Beta diversity	Species Composition	Species Abundance	Phylogenetic Tree
Jaccard index	✓		
Sorensen index	✓		
Bray-Curtis distance	✓	✓	
Euclidean distance	✓	✓	
Weighted UniFrac distance	✓	✓	✓

Taxonomy distribution



[Home](#) [SILVAngs](#) [Browser](#) [Search](#) [ACT](#) [Download](#) [Documentation](#) [Projects](#) [Contact](#)

SILVA

Welcome to the SILVA rRNA database project

A comprehensive on-line resource for quality checked and aligned ribosomal RNA sequence data.

SILVA provides comprehensive, quality checked and regularly updated datasets of aligned small (16S/18S, SSU) and large subunit (23S/28S, LSU) ribosomal RNA (rRNA) sequences for all three domains of life (*Bacteria*, *Archaea* and *Eukarya*).

SILVA are the official databases of the software package ARB.

For more background information → [Click here](#)

SILVA Alignment, Classification and Tree (ACT) Service

The SILVA ACT service combines alignment, search and classify as well as reconstruction of trees in a single web application.

SILVA ACT is available at: → www.arb-silva.de/act



SILVAngs



Check out our service for Next Generation Amplicon data

News

30.10.2024

DSMZ Digital Diversity Workshop



we are happy to announce that this year's DSMZ Digital Diversity Workshop will take place on 11 November 2024, 2 - 4 PM CET online!

11.07.2024

SILVA Release 138.2



The SILVA taxonomy release 138.2 is a maintenance release, with no additional sequences added to the ARB databases since the previous release 138.1. In this update we have focused on Prokaryotes.

11.12.2023

SILVA named Global Core Biodata Resource (GCBR)



SILVA is one of fifteen biodata resources worldwide that have been selected in the Global Biodata Coalition's 2023 selection round for Global Core Biodata Resources (GCBRs), increasing the total number of GCBRs to 52. SILVA joins its DSMZ sister databases *BacDive* and *BRENDA* (both selected in 2022) as well as *LPSN* (2023) as GCBR.

07.12.2023

We are hiring!



SILVA is looking for computer scientists as backend and web developer. Our colleagues from the DSMZ Digital Diversity integration team are also hiring for multiple positions. Find all open positions in our teams [here](#).

[go to Archive ->](#)

<https://www.arb-silva.de>

16S Data Structure

```
>rar <- readRDS("fig1_24line.rds")
```

```
> rar
```

phyloseq-class experiment-level object

otu_table() OTU Table: [1787 taxa and 204 samples]

sample_data() Sample Data: [204 samples by 5 sample variables]

tax_table() Taxonomy Table: [1787 taxa by 6 taxonomic ranks]

```
>sample_data(rar)
```

```
>otu_table(rar)
```

```
>tax_table(rar)
```

```
> rar2 <- readRDS("fig3_200line.rds")
```

```
rar2
```

phyloseq-class experiment-level object

otu_table() OTU Table: [1189 taxa and 598 samples]

sample_data() Sample Data: [598 samples by 16 sample variables]

tax_table() Taxonomy Table: [1189 taxa by 7 taxonomic ranks]

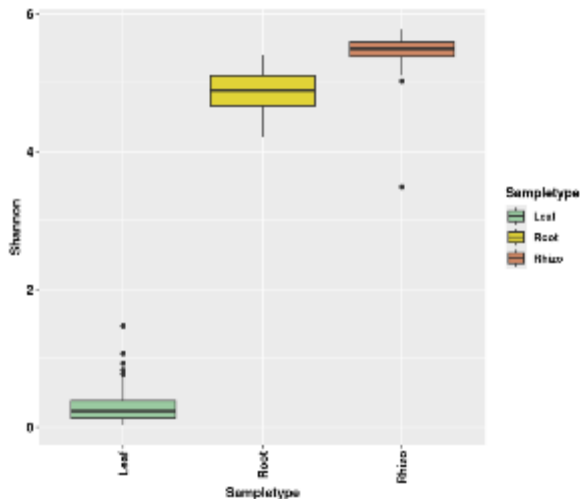
phy_tree() Phylogenetic Tree: [1189 tips and 1188 internal nodes]

α diversity -shannon

```
rar <- readRDS("fig1_24line.rds")
Diversity_table <- estimate_richness(rar, measures=c("Shannon", "Observed"))
Diversity_table <- merge(Diversity_table, sample_data(rar), by="row.names")
Diversity_table$Sampletype <- factor(Diversity_table$Sampletype, levels = c("Leaf", "Root", "Rhizo"))
```

Plot Figure 1b and Figure 1c

```
```{r}
p1b <- ggplot(data=Diversity_table, aes(y=Shannon,x=Sampletype,fill=Sampletype)) +
 geom_boxplot() +
 scale_fill_manual(values=c("#9ACAA1", "#E1D337", "#CE8764")) +
 theme(axis.text.x=element_text(hjust=1,vjust=0.5,size=10,color="black",angle=90,face="bold"),
 axis.text.y=element_text(size=11,color="black",face="bold"),
 axis.title=element_text(size=11,face="bold"),text=element_text(size=11,face="bold"),
 legend.position="right")
p1b
```

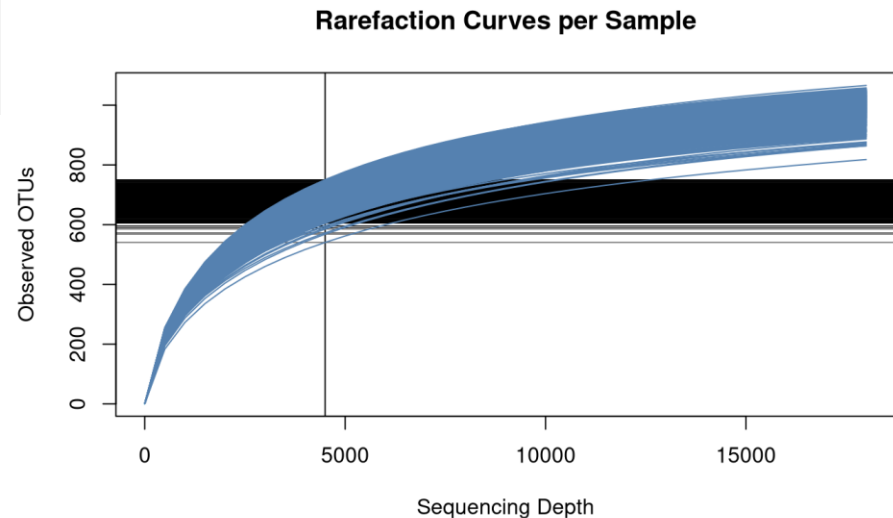


Deng et al., 2021

# Rarefaction analysis

```
rar1 <- readRDS("fig3_200line.rds")
otu_mat <- as(otu_table(rar1), "matrix")
if (taxa_are_rows(rar)) {
 otu_mat <- t(otu_mat)
}

rarecurve(otu_mat,
 step = 500,
 sample = depth,
 col = "steelblue",
 cex = 0.5,
 label = FALSE,
 ylab = "Observed OTUs",
 xlab = "Sequencing Depth",
 main = "Rarefaction Curves per Sample")
```

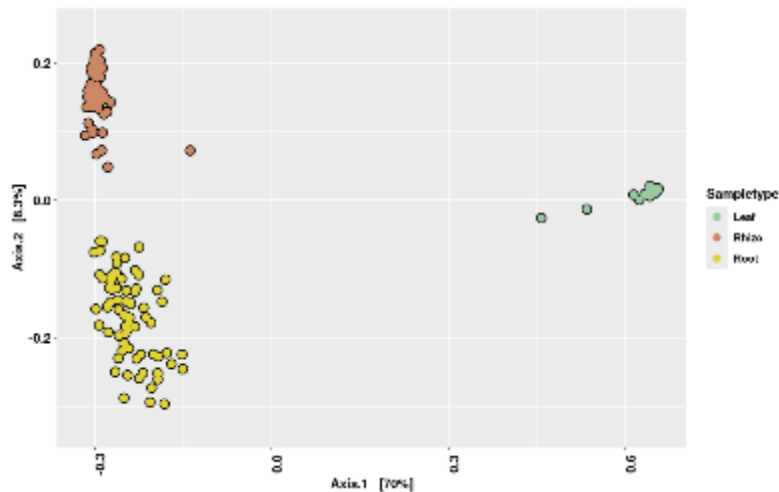


Deng et al., 2021

# $\beta$ diversity - Bray-Curtis distance

```
p1c <- plot_ordination(rar, ordinate(rar, "MDS", distance="bray"), axes=1:2, color = "Sampletype") +
 scale_color_manual(values=c("#9ACAA1", "#CE8764", "#E1D337")) +
 geom_point(colour="black", size=3.5) +
 geom_point(size = 2.5) +
 xlim(-0.4, 0.7) +
 scale_x_continuous(breaks=c(-0.3, 0, 0.3, 0.6)) +
 ylim(-0.33, 0.25) +
 theme(axis.text.x=element_text(size=11, color="black", angle=90),
 axis.text.y=element_text(size=11, color="black"),
 axis.title=element_text(size=11, face="bold"),
 text=element_text(size=11, face="bold"),
 legend.position="right")
```

p1c



Deng et al., 2021

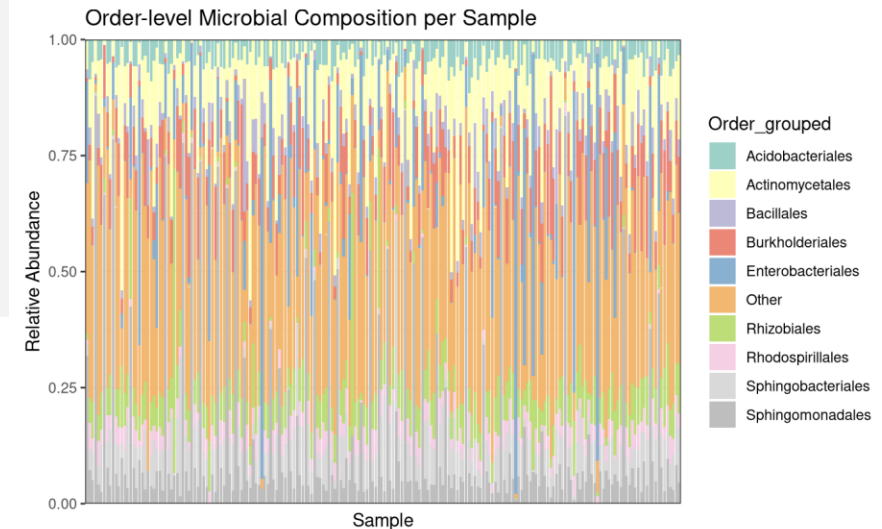
# Taxonomy distribution

```
rar_order <- tax_glom(rar, taxrank = "Order")
compute Order abundance for each sample
rar_order_rel <- transform_sample_counts(rar_order, function(x) x / sum(x))
change phyloseq to dataframe
df_order <- psmelt(rar_order_rel)
set NA as "Unclassified"

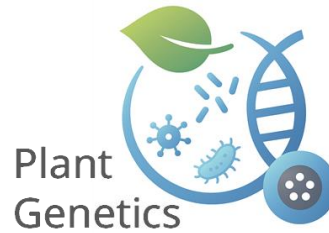
top10 Orders and "Other"
top_orders <- df_order %>%
 group_by(Order) %>%
 summarise(Abundance = sum(Abundance)) %>%
 top_n(9, Abundance) %>%
 pull(Order)

df_order$Order_grouped <- ifelse(df_order$Order %in% top_orders, df_order$Order, "Other")

plot
ggplot(df_order, aes(x = Sample, y = Abundance, fill = Order_grouped)) +
 geom_bar(stat = "identity") +
 theme_bw() +
 labs(title = "Order-level Microbial Composition per Sample",
 x = "Sample",
 y = "Relative Abundance") +
 theme(axis.text.x = element_blank(),
 axis.ticks.x = element_blank(),
 legend.position = "right") +
 scale_y_continuous(limits = c(0, 1), expand = c(0, 0)) +
 scale_fill_manual(values = c(RColorBrewer::brewer.pal(9, "Set3"), "grey"))
```



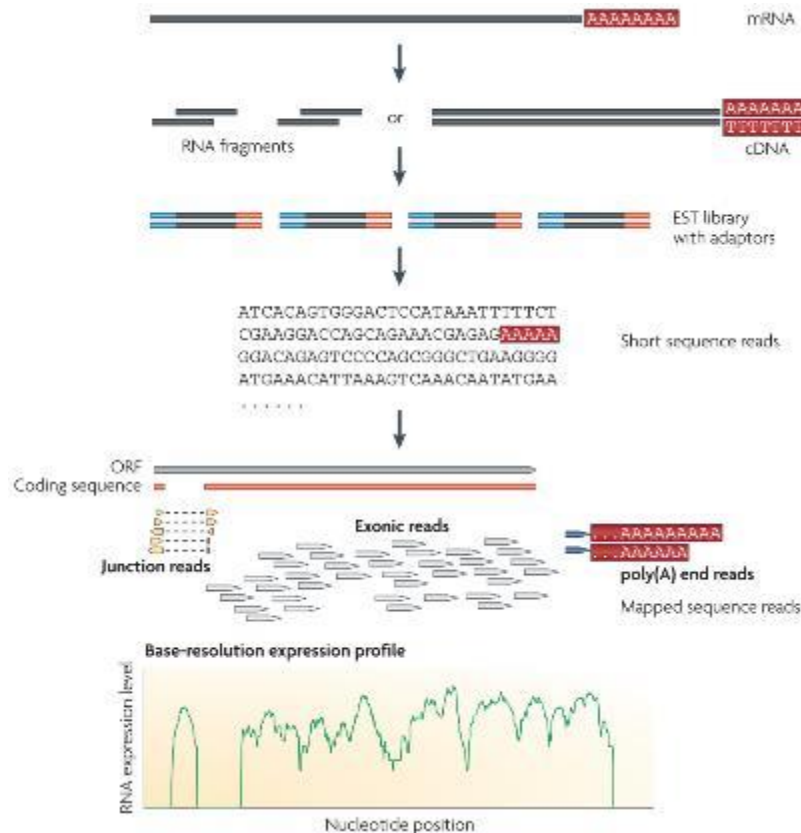
Deng et al., 2021



## III Root transcriptome data analysis (12.8)

# Transcriptome Sequencing

- **Research objective:** To understand the standard RNA-seq data analysis workflow and construct an expression matrix.



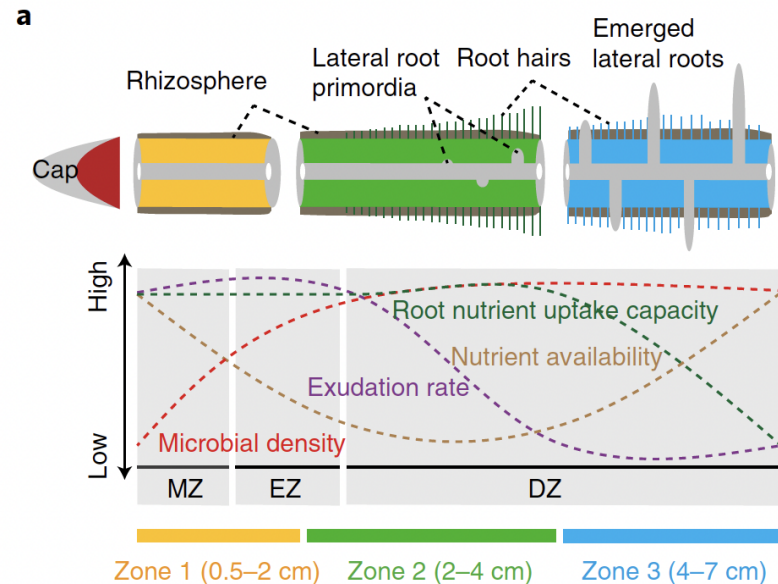
- **Core Applications of RNA-seq**

- Quantifying gene expression levels
- Detecting alternative splicing events
- Assembling transcripts (de novo or reference-guided)
- Detecting gene fusions

(Wang et al., 2009, Nature Reviews Genetics)

# Sample information

30 Maize inbred lines



(Yu et al., 2021, Nature Plants)

Run	Sample name	Description
SRR11839780	B73-Z1-1	zone1 of crown root
SRR11839312	B73-Z1-2	zone1 of crown root
SRR11839301	B73-Z1-3	zone1 of crown root
SRR11839289	B73-Z2-1	zone2 of crown root
SRR11839278	B73-Z2-2	zone2 of crown root
SRR11839508	B73-Z2-3	zone2 of crown root
SRR11839497	B73-Z3-1	zone3 of crown root
SRR11839630	B73-Z3-2	zone3 of crown root
SRR11839619	B73-Z3-3	zone3 of crown root

LH38
LH39
A9
787
IB014
PHN11
PHJ75
PHM10
LH205
PHW51
<b>B73</b>
A632
Mo17
LH51
LH52
E8501
LH93
PHJ31
PHW20
W64A

# RNA sequencing data analysis pipeline

---

## 1. Understanding Raw Sequencing Data

Check read quality using FastQC or MultiQC.

## 2. Read Trimming & Alignment / Quantification

Trim adapters and low-quality reads, then align reads to a genome (STAR, HISAT2).

## 3. Gene / Transcript Quantification

Generate expression matrices (e.g., counts, TPM, FPKM).

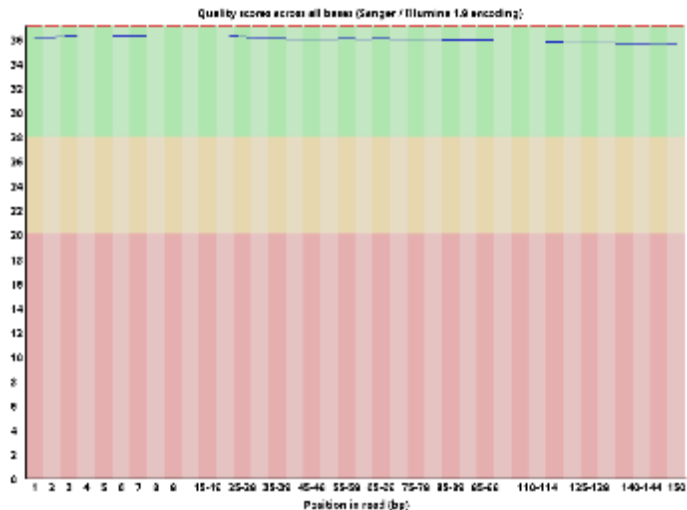
## 4. Data Visualization & Interpretation

Visualize patterns using PCA, heatmaps, volcano plots, etc.

# 1. Understanding Raw Sequencing Data

## Raw reads

```
[go82tip@tuwzz7x-odin RNAseq]$ head SRR11839278_1.fastq
@SRR11839278.1 1 length=150
TCTCACCCTTGTCTCGCGCAACAGCAGTCGGCCGCGGGCCCGGGATCGCTGGCGGGTTGTGCGGGGTTGTGCTCACCCTGCTACTAGCC
GAGCGAGAGAGAGAGAGAGAAG
+SRR11839278.1 1 length=150
FF:F,FFFF,F,F:FFFFFF,F:FF,F:FFF,F,,FFFFFF,,F,:FFFFFFFFFFFF,FFFF,FF,,F,FF:F:FFFFFFF:,F,,FF,
F,F,F,F:FFF,F:FFF:FF,F
@SRR11839278.2 2 length=150
CAGGTCCTAAAATATGGTTTCTCGCAATCAAATACTTCATTCTTGGATGCCCCGGCGCCTACTACCTTTGGTCCCGGCCACTATACCGTG
TTCCTATTCTACTTGGTTCACA
+SRR11839278.2 2 length=150
FFFFFFFF::FFFFFFFF:FF,FFFFFF:,F,,FFFFF,F:::FF,FFFFFFFFFFFF,FFFFFFFF,FF:F:::,::,FFFFFF:F:FFFFFF:F
FFFFF,FFFFFFFFF:,::FFFF
```



## 2. Alignment / Quantification

### Bam file

```
(bioenv) [go82tip@tuwzz7x-odin RNAseq]$ samtools view SRR11839278.uniq.bam | less -S
SRR11839278.15888007 99 chr1 21632 60 18S9M22508N123M = 45636 375 GGAGACCTGGAGGAGACGCTGGAGGGG
SRR11839278.9766371 113 chr1 34648 60 150M = 34793 275 CGGGAACGGAACAAAGACACGCGGCGTGCGGGGCC
SRR11839278.23144857 163 chr1 34659 60 150M = 34850 341 CAAAGACACGCGGCGTGCGGGGCCCTCGCTCCTCTG
SRR11839278.1506825 99 chr1 34667 60 150M = 34750 233 CGCGGCGTGCGGGGCCCTCGCTCCTCTGCTTCCCT
```

position 150 match

### Mapping result

```
[go82tip@tuwzz7x-odin RNAseq]$ more SRR11839278.summary
26469714 reads; of these:
 26469714 (100.00%) were paired; of these:
 2116811 (8.00%) aligned concordantly 0 times
 23712176 (89.58%) aligned concordantly exactly 1 time
 640727 (2.42%) aligned concordantly >1 times

 2116811 pairs aligned concordantly 0 times; of these:
 163073 (7.70%) aligned discordantly 1 time

 1953738 pairs aligned 0 times concordantly or discordantly; of these:
 3907476 mates make up the pairs; of these:
 2389423 (61.15%) aligned 0 times
 1473075 (37.70%) aligned exactly 1 time
 44978 (1.15%) aligned >1 times
95.49% overall alignment rate
```

# 3. Gene / Transcript Quantification

```
(bioenv) [go82tip@tuwzz7x-odin RNAseq]$ head stringtie_uniq/B73-Z2-2/SRR11839278.abun
```

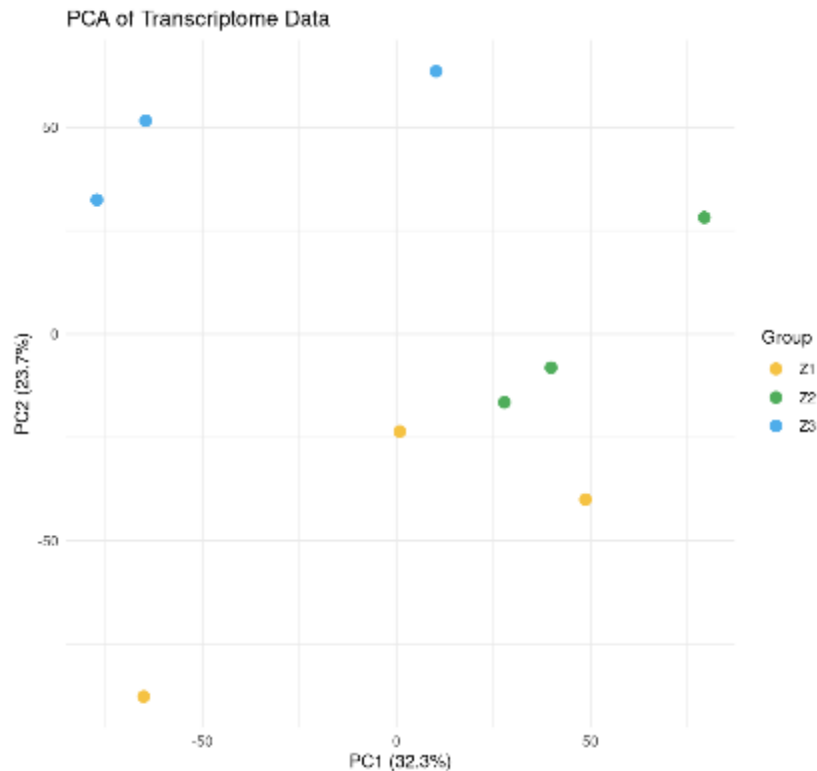
Gene ID	Gene Name	Reference	Strand	Start	End	Coverage	FPKM	TPM
Zm00001eb000150	-	chr1	+	328087	328525	1.637813	0.241440	0.423117
Zm00001eb000170	-	chr1	-	550834	564766	34.270252	5.897928	10.335961
Zm00001eb000180	-	chr1	+	551078	552061	0.000000	0.000000	0.000000
Zm00001eb000010	-	chr1	+	34617	40204	9.302351	1.371314	2.403192
Zm00001eb000020	-	chr1	-	41214	46762	117.030136	22.718590	39.813721
Zm00001eb000050	-	chr1	-	108554	114382	0.000000	0.000000	0.000000
Zm00001eb000060	-	chr1	-	188559	189581	113.641823	16.752607	29.358500
Zm00001eb000070	-	chr1	-	190192	198832	0.143411	0.021141	0.037049
Zm00001eb000080	-	chr1	-	200262	203393	128.814728	18.989334	33.278301

FPKM (Fragments Per Kilobase of transcript per Million mapped reads)

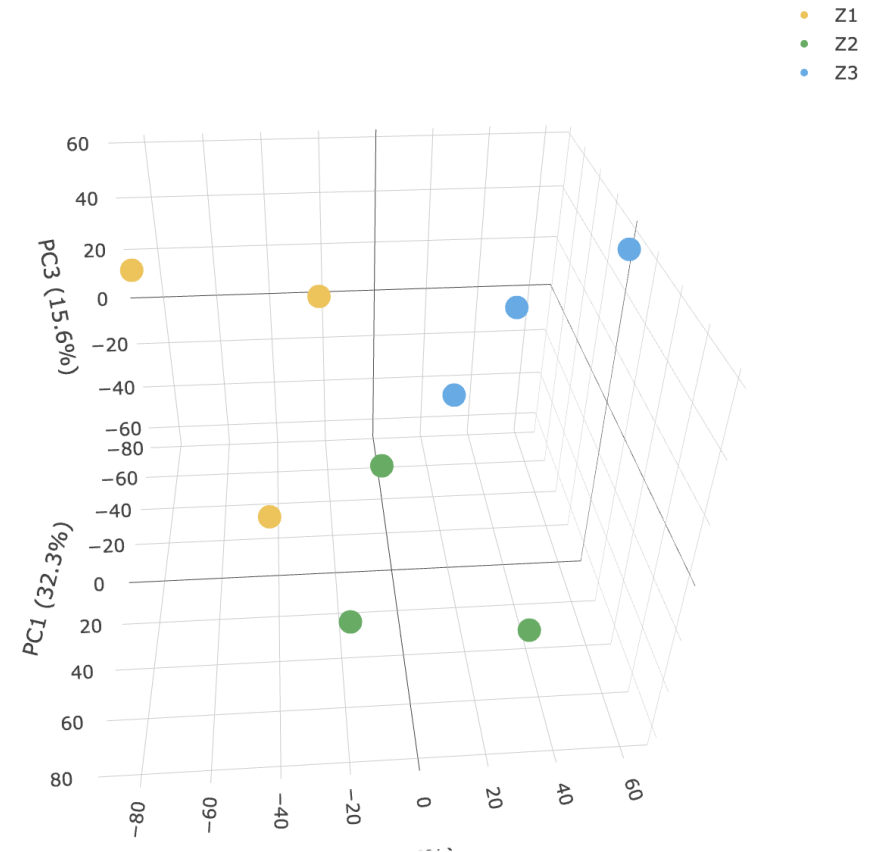
TPM (Transcripts Per Million)

# 4. Data Visualization & Interpretation

## 2D-PCA

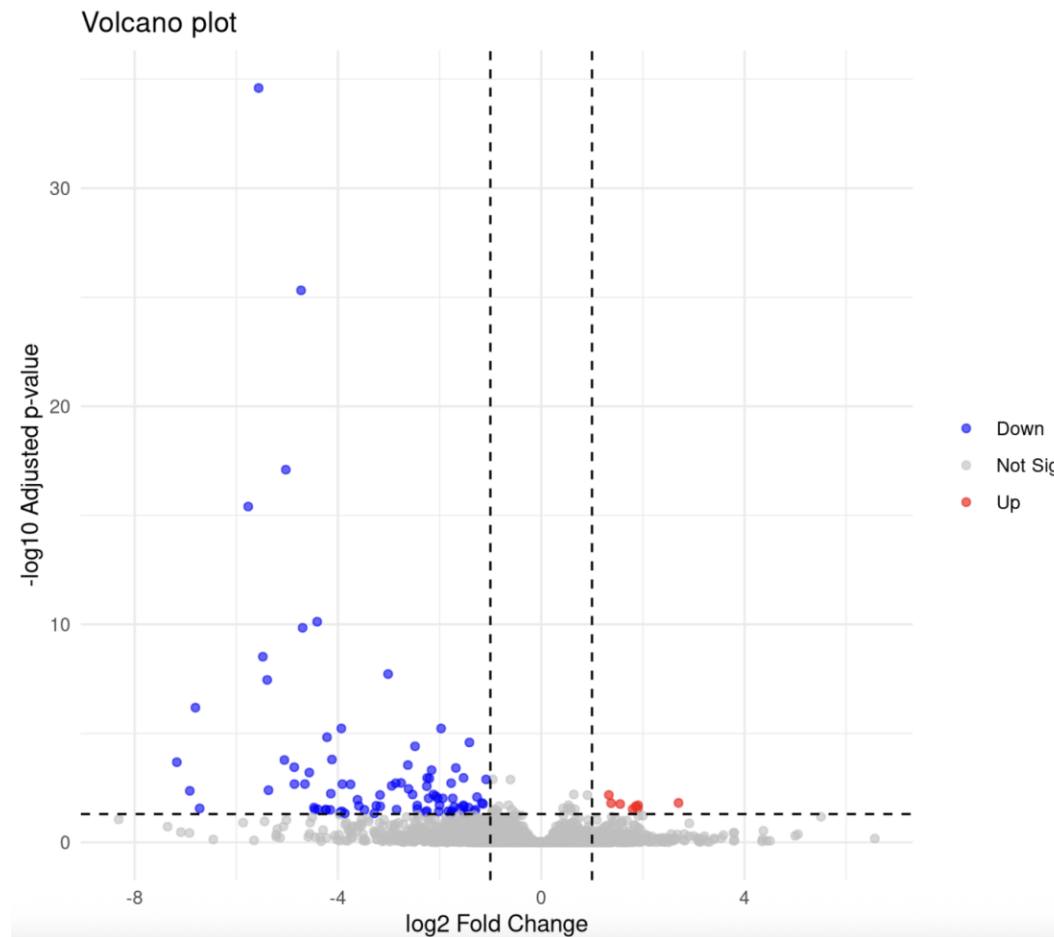


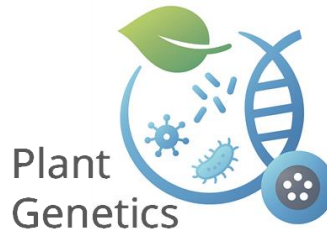
## 3D-PCA



# 4. Data Visualization & Interpretation

## Differential expression analysis

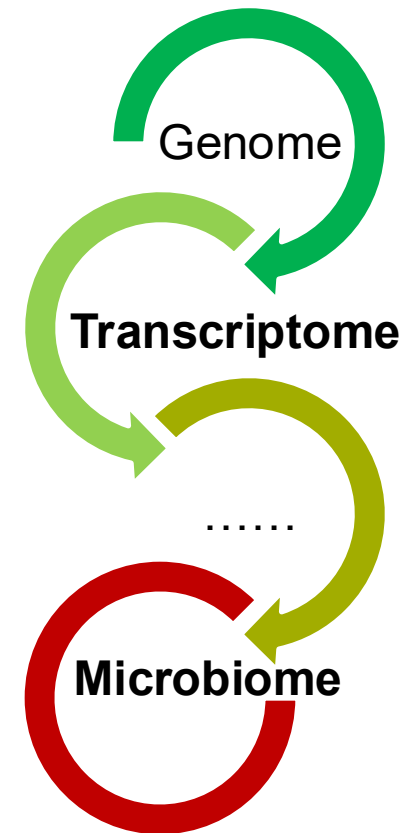
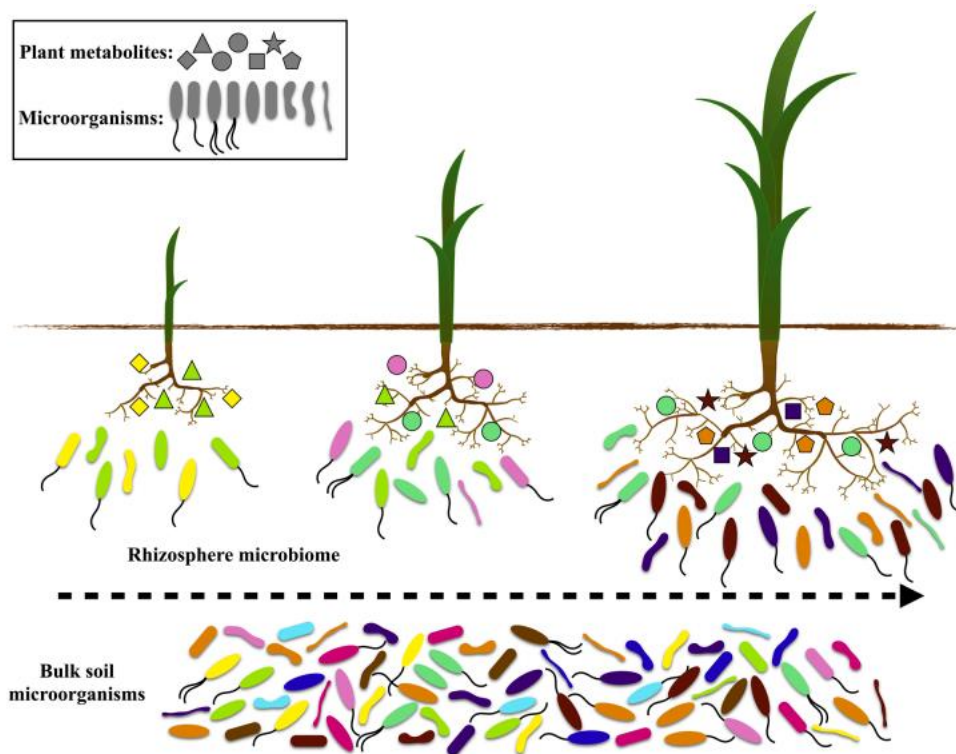




# IV Gene Expression – Microbiome Co-regulation Networks (12.15)

# Root and Microbiome interactions

- **Research objective:** To identify the regulatory relationships between specific gene expression and the abundance of specific microbes.



(Zhang et al., 2021, Microbiological Research)

# Microbiome co-regulation Networks

- Expression matrix

	sample1	sample2
Gene1	FPKM1	FPKM2
Gene2	FPKM3	FPKM4
Gene3	FPKM5	FPKM6
...	...	...

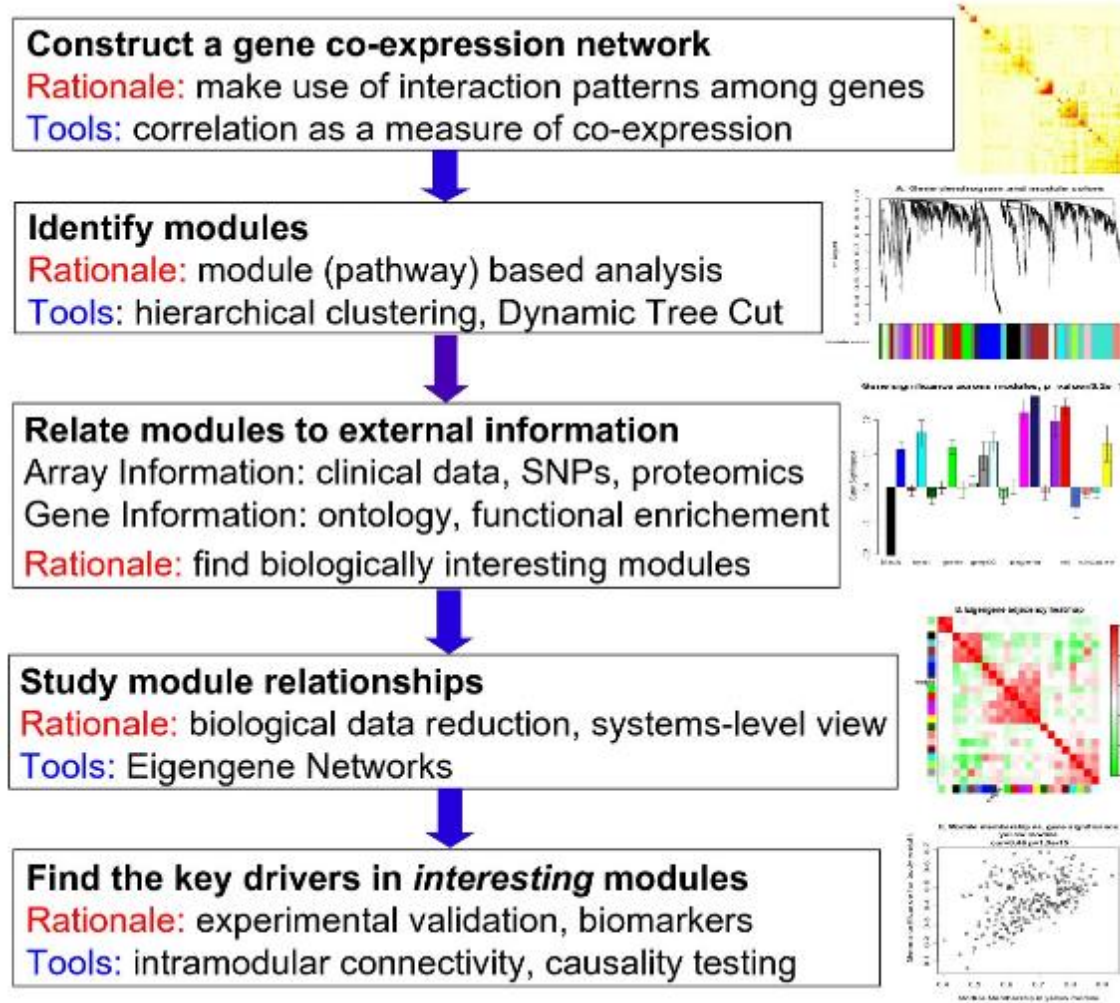
- Microbiome ASV feature table

	sample1	sample2
class1	count1	count2
class2	count3	count4
class3	count5	count6
...	...	...



- Code
- [https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/03\\_Co-regulation Networks/](https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/03_Co-regulation%20Networks/)

# Microbiome Co-regulation Networks



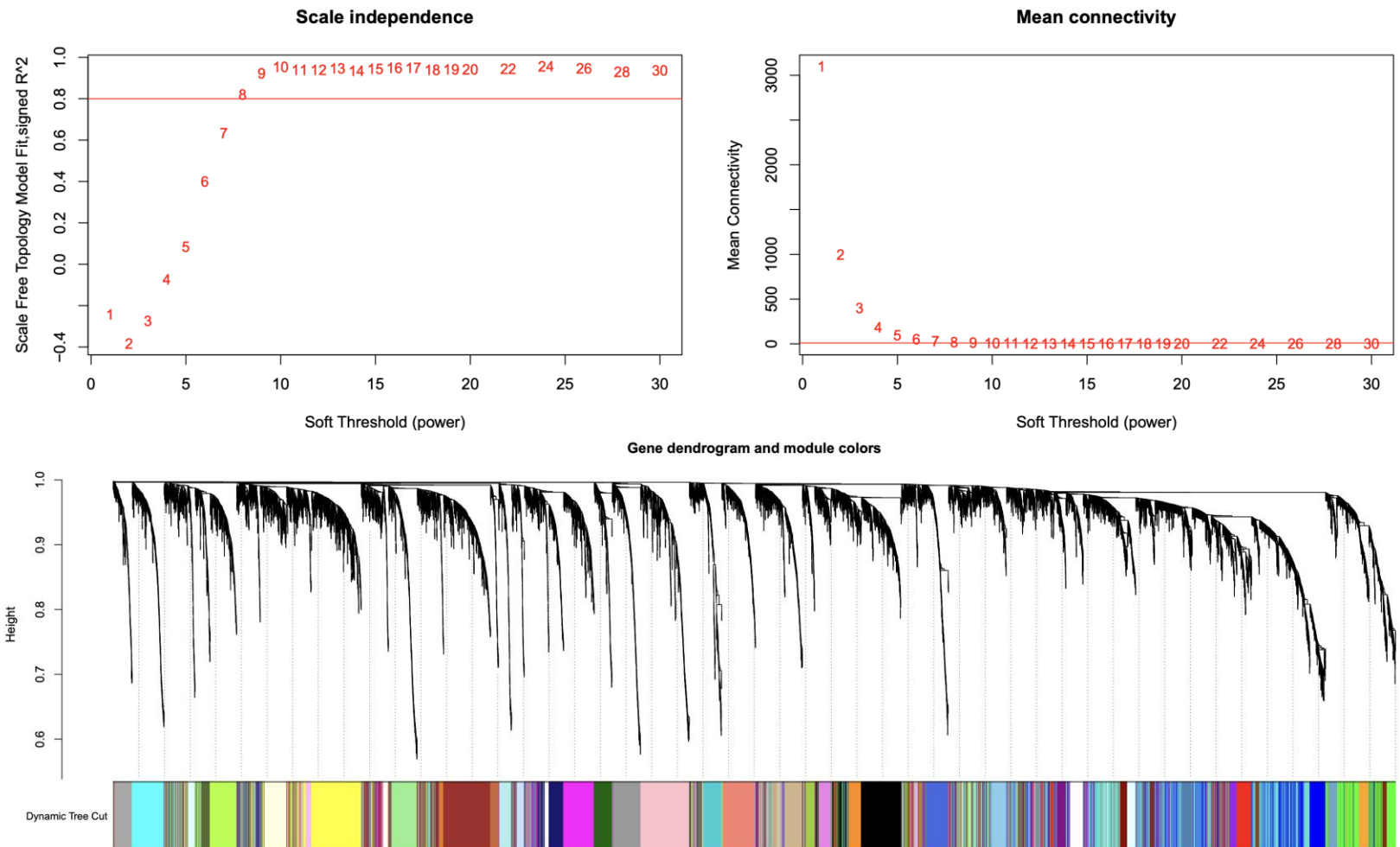
(Peter Langfelder, et al., 2008, Genome Biology)



**Plant  
Genetics**



# Identify modules



How many gene modules have you obtained?

# Relate modules to external information

---

Please use one of the gene modules on the following website to perform the functional enrichment analysis.

Select on gene module (less than 100 genes)



Convert the gene ID from v4 to v5

Based on the convet file from github

[B73v4\\_to\\_B73v5.tsv](#)



Conduct GO enrichment analysis

[https://v2.arabidopsis.org/tools/go\\_term\\_enrichment.jsp](https://v2.arabidopsis.org/tools/go_term_enrichment.jsp)

# Relate modules to external information

Please use one of the gene modules on the following website to perform the functional enrichment analysis.

agriGO

**agriGO v2** GO Analysis Toolkit and Database for Agricultural Community

HOME ANALYSIS SPECIES CUSTOM TOOL DOWNLOAD STATISTIC MANUAL FAQ

**Zea mays Singular Enrichment Analysis (SEA)**

1. Select analysis tool:  
Singular Enrichment Analysis (SEA) ✓  
Parametric Analysis of Gene Set Enrichment (PAGE)  
searching for genes or GO terms

2. Input a gene list:

[zma]

[add]

3. Select reference:  
Suggested backgrounds  
Maize v4 (Ensembl) (eg. Zm000014039580)  
Customized reference [ Example ]

For each species, suggested backgrounds are provided. These backgrounds are all pre-computed, and are available to download. To those species without a relatively completed GO profile, backgrounds from near organisms are used as suggestion. If you don't like these backgrounds, then you may submit your customized with/without GO annotation.

PlantGSAD

**PlantGSAD** The Plant GeneSet Annotation Database

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**Analysis Gene Sets**

Users can submit a list of gene sets for overlap computing.

**Choose Gene Sets**

☐ G1: Gene ontology gene sets

☒ BP: GO biological process  
☒ CC: GO cellular component  
☒ MF: GO molecular function

☐ G2: Other ontology gene sets

☐ PO: Plant Ontology gene sets  
☐ TO: Plant Trait Ontology gene sets  
☐ PECO: Plant Experimental Conditions Ontology gene sets

☐ G3: Pathway gene sets

☒ PlantCyc: PlantCyc gene sets  
☐ KEGG: KEGG gene sets  
☒ Mapman: Mapman gene sets

☐ G4: Gene family based gene sets

☐ TR: Transcription Regulators/Factors  
☐ CAZy: Carbohydrate-Active Enzymes  
☐ PK: Protein Kinase  
☐ Ub: Ubiquitins  
☐ M450: Cytochrome P450  
☐ EAR: EAR motif  
☐ Lip: Lipid metabolism Enzymes  
☐ Car: Chromatin associated factor  
☐ OTH: Other gene family based gene sets

☐ G5: Chromatin states based gene sets

☐ Sta: Chromatin states  
☒ ERG: Epigenetic mark related genes  
☐ CRG: Chromatin associated factor related genes  
☐ HIC: Hi-C identified genes

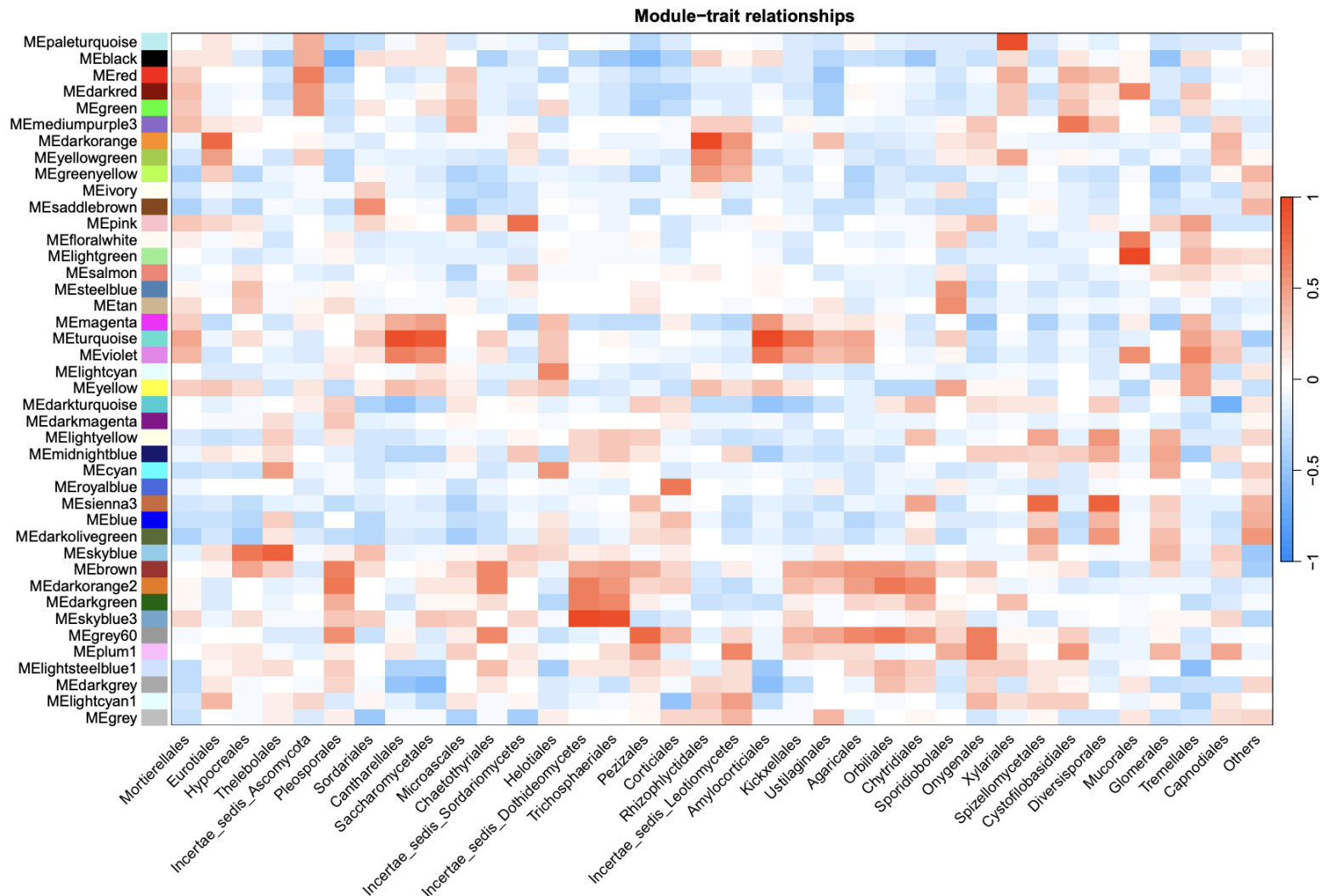
☐ G6: Target gene sets

☒ TFT: Transcription Factor Targets  
☐ MIR: MicroRNA Targets

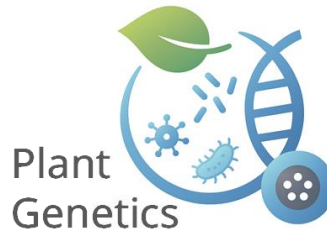
[https://systemsbiology.cau.edu.cn/agriGOv2/specises\\_analysis.php?SpeciseID=20&latin=Zea\\_mays](https://systemsbiology.cau.edu.cn/agriGOv2/specises_analysis.php?SpeciseID=20&latin=Zea_mays)

<https://systemsbiology.cau.edu.cn/PlantGSEAv2/analysis.php>

# Microbiome Co-regulation Networks



- 
- Could you try constructing co-expression networks using different transcriptome datasets or alternative microbial classifications?



**S1 safety instruction !!!**

Thank you

Lecturer Assistant  
Tian Tian